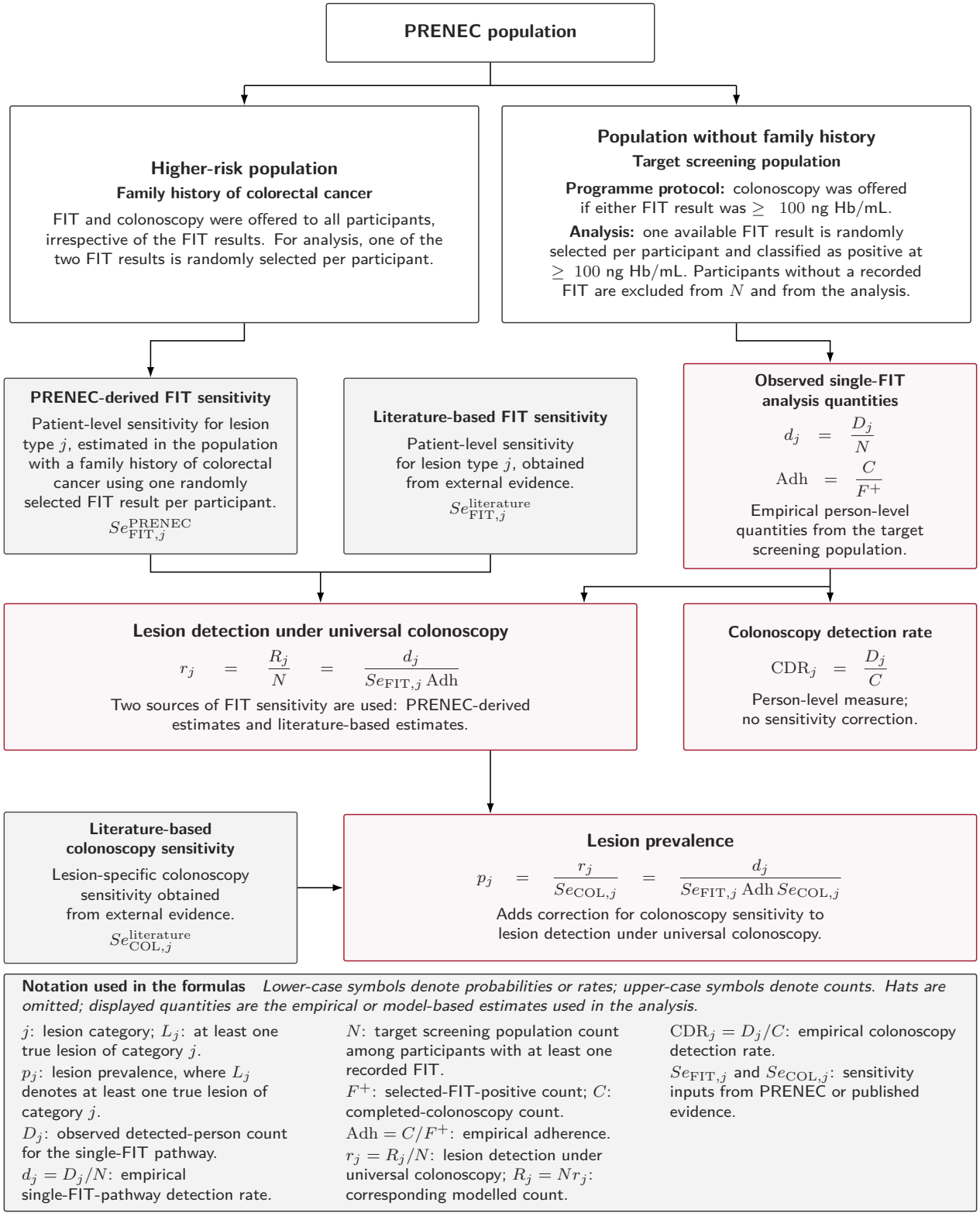
PRENEC population
Higher-risk population
Family history of colorectal cancer
FIT and colonoscopy were offered to all participants, irrespective of the FIT results. For analysis, one of the two FIT results is randomly selected per participant.
Population without family history
Target screening population
Programme protocol: colonoscopy was offered if either FIT result was ≥ 100 ng Hb/mL.
Analysis: one available FIT result is randomly selected per participant and classified as positive at ≥ 100 ng Hb/mL. Participants without a recorded FIT are excluded from N and from the analysis.
PRENEC-derived FIT sensitivity
Patient-level sensitivity for lesion type j, estimated in the population with a family history of colorectal cancer using one randomly selected FIT result per participant.
$Se_{\mathrm{FIT},j}^{\mathrm{PRENEC}}$
Literature-based FIT sensitivity
Patient-level sensitivity for lesion type j, obtained from external evidence.
$Se_{\mathrm{FIT},j}^{\mathrm{literature}}$
Observed single-FIT analysis quantities
$d_j = \frac{D_j}{N}$
$\mathrm{Adh} = \frac{C}{F^+}$
Empirical person-level quantities from the target screening population.
Lesion detection under universal colonoscopy
$r_j = \frac{R_j}{N} = \frac{d_j}{Se_{\mathrm{FIT},j}\,\mathrm{Adh}}$
Two sources of FIT sensitivity are used: PRENEC-derived estimates and literature-based estimates.
Colonoscopy detection rate
$\mathrm{CDR}_j = \frac{D_j}{C}$
Person-level measure; no sensitivity correction.
Literature-based colonoscopy sensitivity
Lesion-specific colonoscopy sensitivity obtained from external evidence.
$Se_{\mathrm{COL},j}^{\mathrm{literature}}$
Lesion prevalence
$p_j = \frac{r_j}{Se_{\mathrm{COL},j}} = \frac{d_j}{Se_{\mathrm{FIT},j}\,\mathrm{Adh}\,Se_{\mathrm{COL},j}}$
Adds correction for colonoscopy sensitivity to lesion detection under universal colonoscopy.
Notation used in the formulas Lower-case symbols denote probabilities or rates; upper-case symbols denote counts. Hats are omitted; displayed quantities are the empirical or model-based estimates used in the analysis.
j: lesion category; L_j: at least one true lesion of category j.
p_j: lesion prevalence, where L_j denotes at least one true lesion of category j.
D_j: observed detected-person count for the single-FIT pathway.
$d_j = D_j/N$: empirical single-FIT-pathway detection rate.
N: target screening population count among participants with at least one recorded FIT.
F^+: selected-FIT-positive count; C: completed-colonoscopy count.
$\mathrm{Adh} = C/F^+$: empirical adherence.
$r_j = R_j/N$: lesion detection under universal colonoscopy; $R_j = Nr_j$: corresponding modelled count.
$\mathrm{CDR}_j = D_j/C$: empirical colonoscopy detection rate.
$Se_{\mathrm{FIT},j}$ and $Se_{\mathrm{COL},j}$: sensitivity inputs from PRENEC or published evidence.